

GANIT MANJARI

Class IX Mathematics

CHAPTER 11

PERIMETER AND AREA

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“Learning Mathematics is Learning to Think.”

Why Should We Study Perimeter and Area?

Motivation

Look around yourself.
Every classroom,
every playground,
every garden,
every book,
every mobile screen,
every house,
and every field has a shape.
To build these objects correctly,
we must answer two important questions.

How long is its boundary?

How much space does it cover?

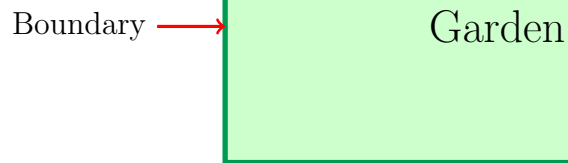
These two questions lead us to one of the most useful chapters of Mathematics—

PERIMETER AND AREA

Imagine This...

Ravi wants to build a beautiful garden in front of his house.
He plans to put a fence around the garden.
Now think carefully.
Should he measure
the inside of the garden,
or
its boundary?

Boundary
 $\Rightarrow$
Perimeter



Did You Know?

Whenever we measure the boundary of a figure,
we are finding its

PERIMETER.

Whenever we measure the region enclosed in-
side a figure,
we are finding its

AREA.

Understanding Perimeter

What is Perimeter?

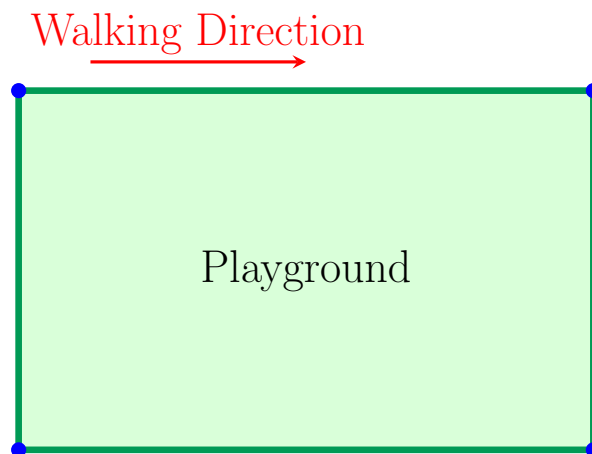
Suppose you start walking from one corner of a playground and keep walking along its boundary until you reach the starting point again.

The total distance travelled is called the

PERIMETER

In simple words,

$$\text{Perimeter} = \text{Total Length of Boundary}$$



Think Before You Answer

A farmer wants to put a fence around his field.
Should he calculate

1. Area
2. Perimeter

Answer : Perimeter

because fencing is always done along the boundary.

Perimeter Around Us

blue!10 Situation	Need
Building a Boundary Wall	Perimeter
Putting Fence Around Garden	Perimeter
Decorating Photo Frame	Perimeter
Border of a Cricket Ground	Perimeter
Border of a Notebook	Perimeter

Easy Memory Trick

Remember the letter

P

P stands for

Police

Police move

around

the city.

Similarly,

Perimeter is always measured

around

the figure.

Activity Time

Take a ruler and measure the boundary of

1. Your Mathematics Book
2. Your Notebook
3. Your Mobile Phone
4. Your Study Table
5. Classroom Door

Can you identify which object has the largest perimeter?

Example–1

A rectangle has
Length = 12 cm
Breadth = 8 cm
Find its perimeter.

Solution

Perimeter of rectangle
 $= 2(l + b)$
 $= 2(12 + 8)$
 $= 2 \times 20$
 $= 40 \text{ cm}$

Amazing Fact

Two different rectangles can have the same perimeter, but their areas may be completely different. You will discover this beautiful fact later in this chapter.

”Perimeter tells us how far we travel around a shape.”

Understanding Area

What is Area?

Suppose Ravi wants to spread green grass inside his rectangular garden.

Now think carefully.

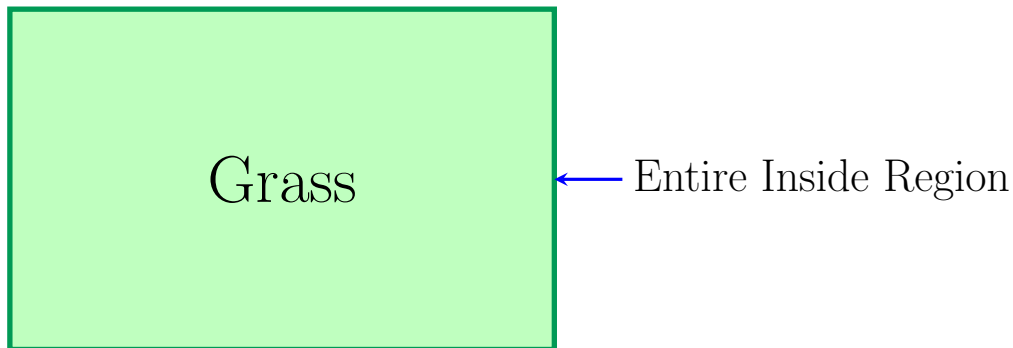
Will he measure the boundary of the garden?

No!

He must measure the entire region enclosed inside the boundary.

This enclosed region is called the

AREA



Memory Trick

Remember the word

AREA

Think of

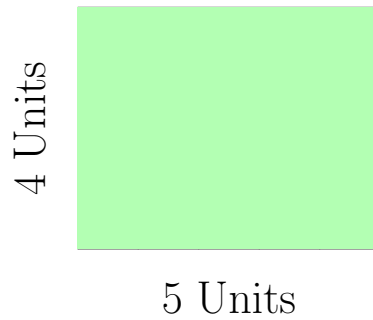
All Region Enclosed Always

Although this is only a memory trick (not the real full form), it helps us remember that

Area always refers to the inside region.

Measuring Area Using Unit Squares

Every area is measured by counting equal-sized square units.



Number of rows = 4

Number of columns = 5

Total unit squares = $4 \times 5 = 20$

Important Conclusion

$$\text{Area} = \text{Length} \times \text{Breadth}$$

This formula is obtained simply by counting unit squares.

Why is $\text{Area} = \text{Length} \times \text{Breadth}$?

Let's Understand Visually

Imagine that a rectangle is completely covered with small unit squares. Instead of counting every square one by one, we can count

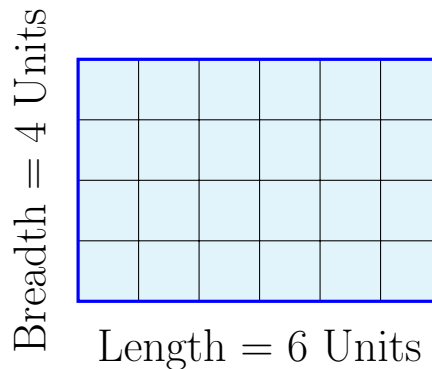
- Number of rows
- Number of columns

Then,

$$\begin{aligned} &\text{Total Squares} \\ &= \\ &\text{Rows} \\ &\times \\ &\text{Columns} \end{aligned}$$

Hence,

$$\text{Area} = \text{Length} \times \text{Breadth}$$



$$\text{Rows} = 4$$

$$\text{Columns} = 6$$

$$\begin{aligned} &\text{Total Squares} \\ &= 4 \times 6 = 24 \end{aligned}$$

Important Note

Area always tells us

How much surface is covered?

It never tells us

How long the boundary is.

Example 2

Find the area of a rectangle whose

Length = 12 cm

Breadth = 7 cm.

Solution

Area

=

Length $\times$ Breadth

= 12×7

84 cm²

Think and Answer

Two rectangles have the same area.

Can their perimeters be different?

☐ Yes

☐ No

Amazing Fact

Two figures may have

Same Area

but

Different Perimeters.

Similarly,

two figures may have

the same perimeter,

but different areas.

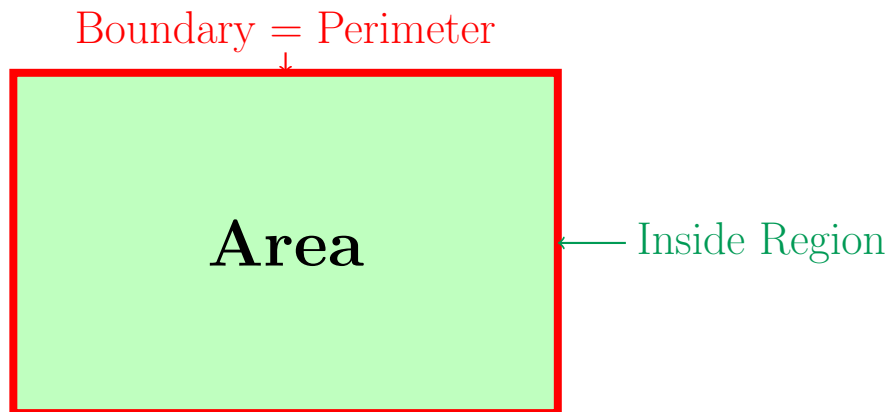
This beautiful idea appears in higher Mathematics too.

**”Understanding is more powerful than
memorising.”**

Perimeter vs Area

Can You Spot the Difference?

Many students confuse **Perimeter** with **Area**. Let us understand the difference through a simple rectangle.



RoyalBlue!20 Perimeter	Area
Distance around a figure	Region enclosed inside a figure
Measured in cm, m, km	Measured in cm^2 , m^2 , km^2
Needed for fencing	Needed for carpeting or painting
Depends on boundary	Depends on surface covered

Real Life Examples

- Fencing a garden $\Rightarrow$ Find **Perimeter**
- Painting a wall $\Rightarrow$ Find **Area**
- Carpet for a room $\Rightarrow$ Find **Area**
- Ribbon around a photo frame $\Rightarrow$ Find **Perimeter**
- Tiles on a floor $\Rightarrow$ Find **Area**

Quick Quiz

Tick the correct answer.

1. A farmer wants to fence his field.

☐ Area

☐ Perimeter

2. A painter wants to paint a wall.

☐ Area

☐ Perimeter

3. A tailor wants to stitch lace around a table cloth.

☐ Area

☐ Perimeter

Common Mistakes

- Never write perimeter in cm^2 .
- Never write area in cm.
- Boundary is not Area.
- Area is not Boundary.
- Always write the correct unit.

Golden Rule

Boundary $\implies$ Perimeter

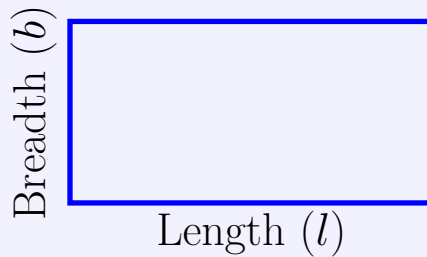
Inside Region $\implies$ Area

**“Remember the Boundary,
Understand the Area.”**

Formula Palace

"Understanding a formula is far more valuable than memorising it."

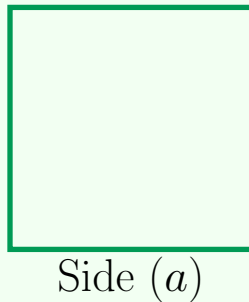
Rectangle



$$\text{Perimeter} = 2(l + b)$$

$$\text{Area} = l \times b$$

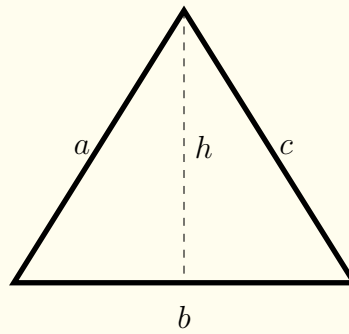
Square



$$\text{Perimeter} = 4a$$

$$\text{Area} = a^2$$

Triangle



$$\text{Perimeter} = a + b + c$$

$$\text{Area} = \frac{1}{2}bh$$

Memory Card

Shape	Formula
Rectangle	$l \times b$
Square	a^2
Triangle	$\frac{1}{2}bh$

Think Like a Mathematician

A rectangle has
 Length = 10 cm
 Breadth = 6 cm
 Without using any calculator,
 can you immediately tell
 its

- Perimeter?
- Area?

**Learn the Concept,
 The Formula Will Never Be Forgotten.**

Explore the Concept

Activity–1 : Same Perimeter

Suppose each rectangle has a perimeter of 20 cm.
Can their areas still be different?
Let us investigate.

Length	Breadth	Perimeter	Area
8	2	20	16
7	3	20	21
6	4	20	24
5	5	20	25

Observation

Although every rectangle has the same perimeter,
their areas are different.
The square gives the maximum area.

Think

Why does a square always enclose the greatest area among all rectangles having the same perimeter?
Discuss with your classmates.

Activity–2 : Draw and Discover

Draw three rectangles on graph paper.

1. 6×2
2. 4×3
3. 5×5

Count the unit squares.
Compare their areas and perimeters.
Write your conclusion.

Fun Fact

Engineers and architects use the concept of Perimeter and Area while designing

- ✓ Houses
- ✓ Parks
- ✓ Stadiums
- ✓ Roads
- ✓ Swimming Pools

Mathematics is everywhere!

Observe → Think → Discover

Solved Examples

“Read the question carefully, understand the concept, then solve step by step.”

Example 1

A rectangular park is 18 m long and 12 m wide.

Find

1. its perimeter,
2. its area.

Solution

Given,

Length = 18 m

Breadth = 12 m

Perimeter

$$= 2(l + b)$$

$$= 2(18 + 12)$$

$$= 2 \times 30$$

$$= 60 \text{ m}$$

Area

$$= l \times b$$

$$= 18 \times 12$$

$$= 216 \text{ m}^2$$

Example 2

The side of a square is 15 cm.
Find

1. Perimeter
2. Area

Solution

Given,
Side = 15 cm

$$\begin{aligned}\text{Perimeter} \\ &= 4 \times 15 \\ &= 60 \text{ cm}\end{aligned}$$

$$\begin{aligned}\text{Area} \\ &= 15^2 \\ &= 225 \\ &= 225 \text{ cm}^2\end{aligned}$$

Example 3

A rectangular classroom is 9 m long and 7 m wide.
Carpet is to be laid on the floor.
How much carpet is required?

Solution

Carpet covers the inside region.
Hence, we calculate the Area.

$$\begin{aligned}\text{Area} \\ &= 9 \times 7 \\ &= 63 \text{ m}^2\end{aligned}$$

Therefore,
63 m² of carpet is required.

Try Yourself

Find the perimeter and area of:

1. A rectangle of length 20 cm and breadth 9 cm.
2. A square of side 18 cm.
3. A rectangle of length 14 m and breadth 5 m.

Teacher's Tip

Always ask yourself:

- Is the question asking about the **boundary**?
Then use **Perimeter**.
- Is the question asking about the **inside region**?
Then use **Area**.

Practice makes Mathematics Beautiful!

Practice Zone

“Practice is the Key to Success.”

A. Multiple Choice Questions

1. The boundary of a closed figure is called
 - (A) Area
 - (B) Volume
 - (C) Perimeter
 - (D) Surface
2. The area of a rectangle of length 9 cm and breadth 5 cm is
 - (A) 14 cm^2
 - (B) 45 cm^2
 - (C) 28 cm^2
 - (D) 90 cm^2
3. Which unit is used for Area?
 - (A) cm
 - (B) m
 - (C) cm^2
 - (D) km

B. Short Answer Questions

1. Define Perimeter.
2. Define Area.
3. Write the formula for the perimeter of a rectangle.
4. Write the formula for the area of a square.
5. Why is area measured in square units?

C. Long Answer Questions

1. A rectangular field is 30 m long and 18 m wide. Find its perimeter and area.
2. A square park has side 24 m. Find its perimeter and area.
3. A classroom is 12 m long and 8 m wide. How many square metres of tiles are required to cover the floor?

D. Higher Order Thinking Skills (HOTS)

1. Can two rectangles have the same area but different perimeters? Give an example.
2. Can two rectangles have the same perimeter but different areas? Explain with an example.
3. Why does a square enclose the maximum area among all rectangles having the same perimeter?

E. Real-Life Application

A gardener wants to put a fence around a rectangular garden of length 16 m and breadth 10 m.

1. How much fencing wire is required?
2. If grass is to be planted inside the garden, what area will be covered?

Challenge Corner

A wire of length 40 cm is bent to form different rectangles.
Which rectangle will have the greatest area?
Explain your answer.

Self Check

Tick the correct statement.

- ☐ I can find the perimeter of a rectangle.
- ☐ I can calculate the area of a square.
- ☐ I know the difference between perimeter and area.
- ☐ I can solve daily-life problems related to perimeter and area.

“Mathematics is everywhere around us.”